Problem Statement

ABC organization is expanding its successful physical store to an online platform. The focus of this case study is to design and implement the User Module of the online shopping application using React.

Design and develop a real-time e-commerce application with the necessary functionalities to view products, add to the required product(s) to the cart, check out, etc.

Functional Requirements

The major focus of this assignment is to build the Front-end aspect of the e-commerce application developed in the last week. This solution will invoke the APIs built so far to implement the requirements added in the subsequent sections.

User Module

1. **User Registration & Login**
   A. **User Registration:** New users can sign up by providing their name, email, and password.
   B. **User Login:** Registered users can log in using their email and password.
2. **Product Browsing & Search**
   A. **Product Listing:** Users can browse all available products.
   B. **Product Search:** Users can search for products by name or category.
   C. **Product Details:** Users can view detailed information about each product.
3. **Cart Management**
   A. **Add to Cart:** Users can add products to their shopping cart.
   B. **View Cart:** Users can view items in their cart, update quantities, and remove items.
   C. **Cart Summary:** Users can see the total amount and item count.
4. **Order Placement**
   A. **Checkout:** Users can proceed to checkout, review their order, and choose a payment method (Credit Card or Cash on Delivery).
   B. **Order History:** Users can view their past orders and order details.
5. **Non-Functional Requirements**
   A. **Security:** Secure Authentication and data protection.
   B. **Performance:** Fast and responsive user experience.
   C. **Scalability:** Designed for future enhancements.

Validation Instructions

Registration Page

Description: A simple form with fields for name, email, password, and a register button.

Validation Instructions:

1. **Name:** Required, minimum 6 characters, letters only.
2. **Email:** Required, must be a valid email format, unique (check against existing users).
3. **Password:** Required, minimum 8 characters, must include at least one uppercase letter, one lowercase letter, one number, and one special character.
4. **Register Button:** Enabled only when all fields are valid. On submission, display error messages for invalid fields and prevent registration if validation fails.

Login Page

Description: A form with email and password fields and a login button.

Validation Instructions:

1. **Email:** Required, must be a valid email format.
2. **Password:** Required.
3. **Login Button:** Enabled only when both fields are valid. On submission, display error messages for invalid credentials or empty fields.

Product Listing Page

Description: A search bar at the top, followed by a grid/list of product cards with name, category, price, and Add to Cart button.

Validation Instructions:

1. **Search Bar:** Optional input, filters products by name or category as user types.
2. **Add to Cart Button:** Enabled only if the product is in stock (Available Quantity > 0). If not, display "Out of Stock" message.
3. **Product Cards:** Display all required product details; ensure no fields are left blank.

Cart Page

Description: A list of cart items with quantity, price, remove button, total amount, and a checkout button.

Validation Instructions:

1. **Quantity:** Must be a positive integer, cannot exceed available stock. If user enters an invalid quantity, show error and prevent update.
2. **Remove Button:** Removes the item from the cart.
3. **Total Amount:** Automatically updates when items are added/removed or quantity changes.
4. **Checkout Button:** Enabled only if cart is not empty.

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Order History Page

Description: A list of past orders with order number, date, total, and a button to view details.

Validation Instructions:

1. **Order List:** Display only orders belonging to the logged-in user.
2. **Order Details:** Must show all products, quantities, and prices for each order.
3. **No Orders:** If no past orders, display a message: "No orders found."

User Interface

Visual Designs

Use React components for Header, Menu, Main Content, and Footer.

Project Layout [Landing Page]: The layout must be designed keeping in mind the following elements for the user interface.

1. **Header Component:**
   A. Display at the top of the page.
   B. Include brand name, site navigation, and profile icon.
2. **Menu Component:**
   A. Positioned ideally on the side or top for navigation purposes.
   B. Contains links or buttons for accessing different sections or pages of the application.
3. **Main Content Component:**
   A. Central area where the primary content of each page is displayed.
   B. Content may vary based on user interactions or navigation.
4. **Footer Component:**
   A. Located at the bottom of the page.
   B. Contains auxiliary links, copyright information, and possibly contact details.

Guidelines

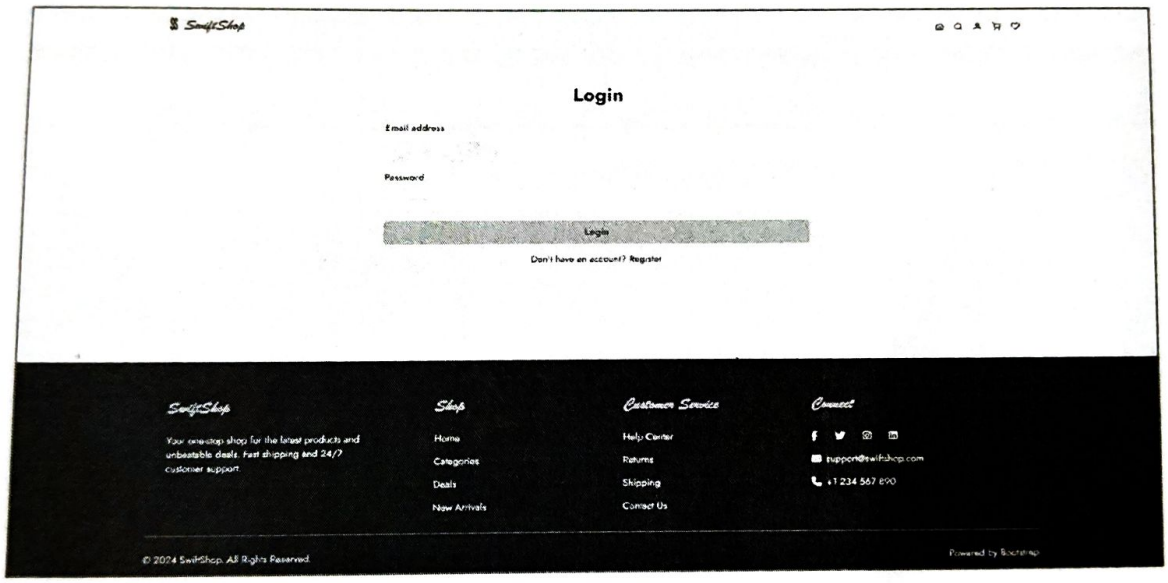
1. **Validations:**
   A. Form with all the expected validations.
   B. Proper validation messages and dropdown should be maintained.
2. **Buttons:** Should have the facility to:
   A. Create a Form:
      I. Enable only when a form is valid.
      II. Disable while API request is in progress.
      III. On Failure, show an error message from server.
   B. Reset the form – All Fields to be reset to blank.

Reference UI Layouts

Register Page



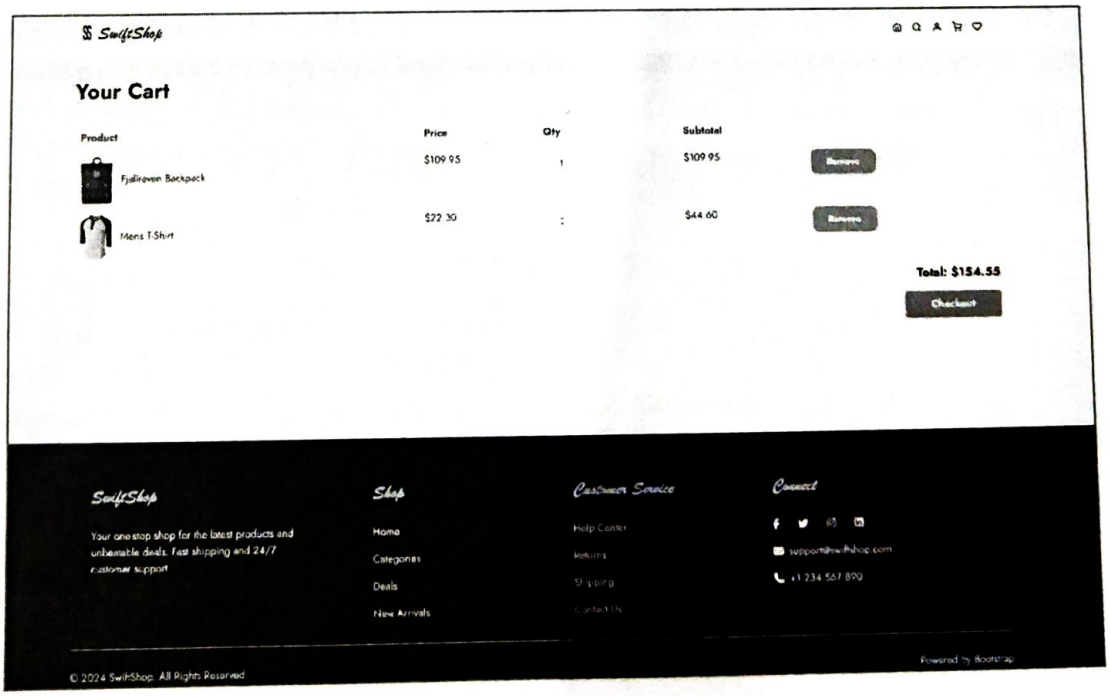
Login Page



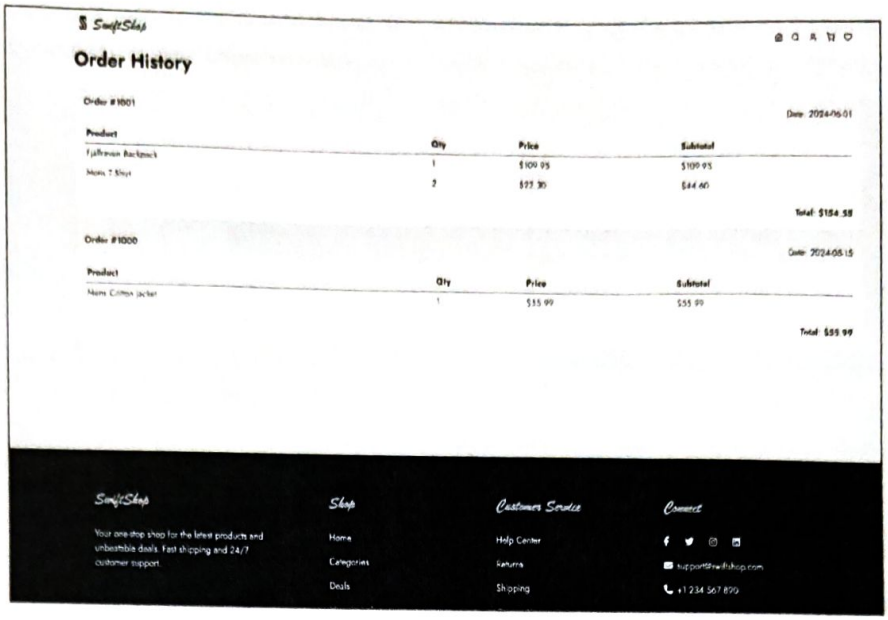
Product Listing



Cart



Order History



Non-Functional Requirements

- The application should have low latency and high throughput.
- The system should be scalable and maintainable.

Best Practices

- Create reusable components.
- Maintain consistent React coding styles and naming conventions.
- Add comments wherever needed.
- Use strict type scripting and avoid 'any'.
- Utilize ES6 features.

Guidelines

- Use best practices for components and service design.
- Add Template Driven/Reactive forms with suitable validation wherever applicable.
- Keep UI simple and user-friendly.
- You can make assumptions and incorporate additional features or functionalities.
- A simple but working prototype is preferred over a nonworking but good-looking application.
- The screenshots provided in this document are only to enable your understanding and can be used as a reference to think through the logic and develop the application.
- Add relevant comments for proper documentation.
- Follow the naming conventions.
- Build reusable components.
- Use Git to maintain project repository, manage pull requests, and resolve conflicts.

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Milestones and Evaluation Criteria

Milestone#	Milestone Description
Milestone 1	User Registration and Login functionality with proper validations.
Milestone 2	Use appropriate coding logic with the required validations to implement product listing and search product components.
Milestone 3	Implement cart management functionality.
Milestone 4	Viewing the order history.
Milestone 5	Implement order placement and order history.
Milestone 6	Push the build to your GitHub repository.
Technical Interview/Viva	Final testing and documentation.